TrES-1b

R-1483

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-1_140605` · created 2026-08-01T13:06:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456813.7988 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.077 +/- 0.05
Transit depth (Rp/Rs)^2	0.59 +/- 0.76 [%]
Orbital Inclination (inc)	85.21 +/- 1.47
Airmass coefficient 1 (a1)	1.22 +/- 0.18
Airmass coefficient 2 (a2)	-0.16 +/- 0.1
Scatter in the residuals of the lightcurve fit is	14.91 %
Best Comparison Star	#1 - [225, 105]
Optimal Aperture	4.28
Optimal Annulus	16.13
Transit Duration (day)	0.049 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.077 ± 0.05 vs 0.1358 (deviation 1.1σ)
Duration fit vs published	0.049 d vs 0.1042 d (deviation 1.57σ)
Mid-time O–C	-3.2 min
Depth SNR	1.4
Residual scatter	14.91 %

Light curve

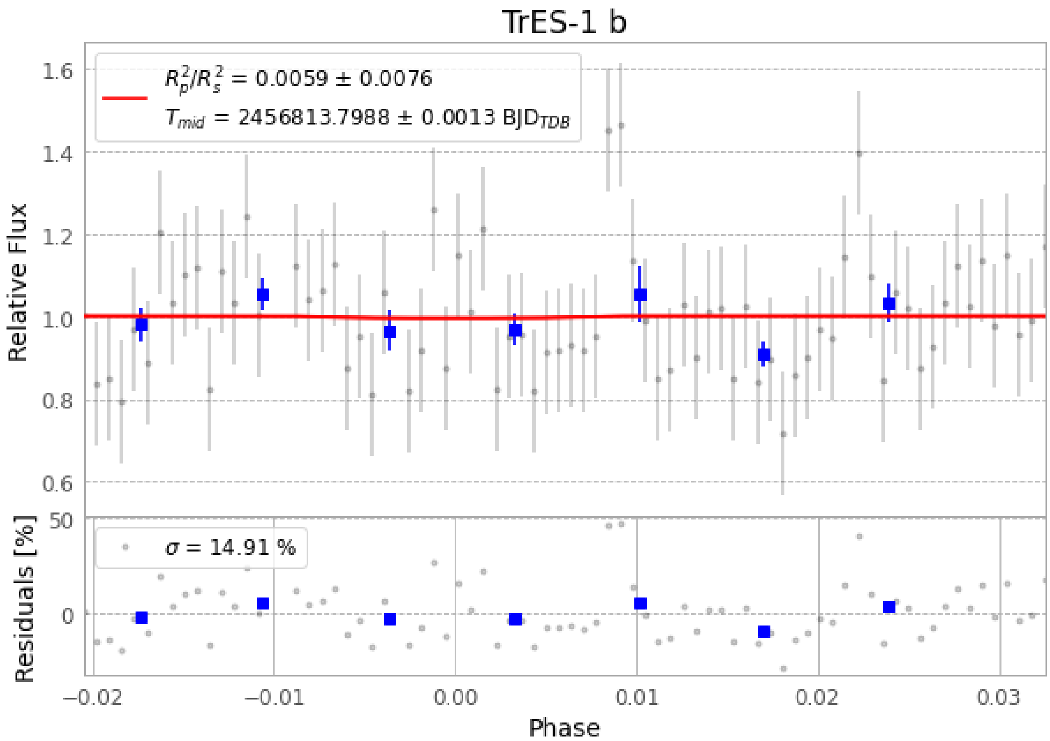

Figure 1 — FinalLightCurve_TrES-1 b_2014-06-04.png (SHA-256 724c026feff05676…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [361, 102], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c6f26b61a83a46c8…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

78 FITS frames (51 MB), `TRES-1140605053612.FITS` → `TRES-1140605092712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-140605.log (9625ea5b905a…), FinalLightCurve_TrES-1 b_2014-06-04.png (724c026feff0…), FinalLightCurve_TrES-1 b_2014-06-04.csv (0f5db4f37849…)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)